

Inherent Circular Dichroism of Phonons in Magnetic Weyl Semimetal $\text{Co}_3\text{Sn}_2\text{S}_2$

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We investigated the infrared-active phonons in ferromagnetic Weyl semimetal $\text{Co}_3\text{Sn}_2\text{S}_2$ using optical spectroscopy. Below the Curie temperature ($T_C \approx 175$ K), we observed asymmetric Fano line shapes of phonons peaks in the optical conductivities, reflecting the presence of electron-phonon coupling. Additionally, the detected phonon signals by the polar Kerr rotation and the ellipticity spectroscopy indicate the circular dichroism (CD) of phonons. We attribute the CD of phonons to their distinct couplings with charge excitations on the tilted Weyl nodal rings in two circularly polarized channels. Our findings provide experimental evidence that, without external fields, phonons can also become circularly polarized by coupling with the electronic topology. Since the magnetic exchange splitting gradually shifts the topological bands in $\text{Co}_3\text{Sn}_2\text{S}_2$, the CD of phonons exhibits significant temperature dependence, hinting at a promising approach for manipulation.

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Introduction—The chirality in bosonic systems has garnered broad interest in condensed matter physics. An intriguing example is the chiral phonon, which arises from the rotational movement of atoms perpendicular to their propagation direction [1–5]. Such circular motion imparts specific angular momentum and magnetic moment to the phonons [6–8], giving rise to Einstein–de Haas effect, the phonon Hall effect, the anomalous thermal Hall effect, etc. [9–18]. In these phenomena, circularly polarized phonons serve as carriers of energy and information [19–25]. Additionally, the circular motion of atoms in magnetic materials couples strongly with the electron spins, which can be used to manipulate the magnetism [26,27]. Despite extensive theoretical investigations, circularly polarized phonons have only been observed in a limited number of materials [3–5,28–30]. Thus, realizing and manipulating circularly polarized phonons in more materials is crucial both for fundamental understanding and practical applications.

In the monolayer of the transition metal chalcogenides [28], circular modes were observed at K and K' with zero propagating velocity; while in chiral crystals like α –HgS [3], quartz [4] and Te [5,25] chiral phonons with huge group velocity can be obtained in the vicinity of Brillouin zone center (Γ point) due to the noncentrosymmetric nature of those materials. In the Dirac semimetal Cd_3As_2 [31] and quantum magnet CoTiO_3 [32], the coupling between phonons and charge excitations results in circularly polarized phonons with exceedingly large magnetic moments [33–37]. However, an external magnetic field is required to break the degeneracy of phonons with opposite polarizations. Whether intrinsic circularly polarized phonons can exist in centrosymmetric systems with spontaneous $\mathcal{T}$ breaking remains an open question. Very recently, theoretical studies predicted that, in systems with broken $\mathcal{T}$, the coupling between phonons and electronic topology could lift the degeneracy of phonons at the Γ point, generating circularly polarized modes even without an external magnetic field [7,8,38,39]. Nevertheless, experiments on such a phenomenon in magnetic topological materials are still pending verification. To reveal the circularly polarized phonons in magnetic topological materials, one needs to determine (i) whether there exists considerable electron-phonon coupling (EPC), and (ii) whether the EPC can result in the circular dichroism (CD) of phonons.

Optical spectroscopy is a powerful tool for detecting the infrared-active phonons around the Γ point. The coupling

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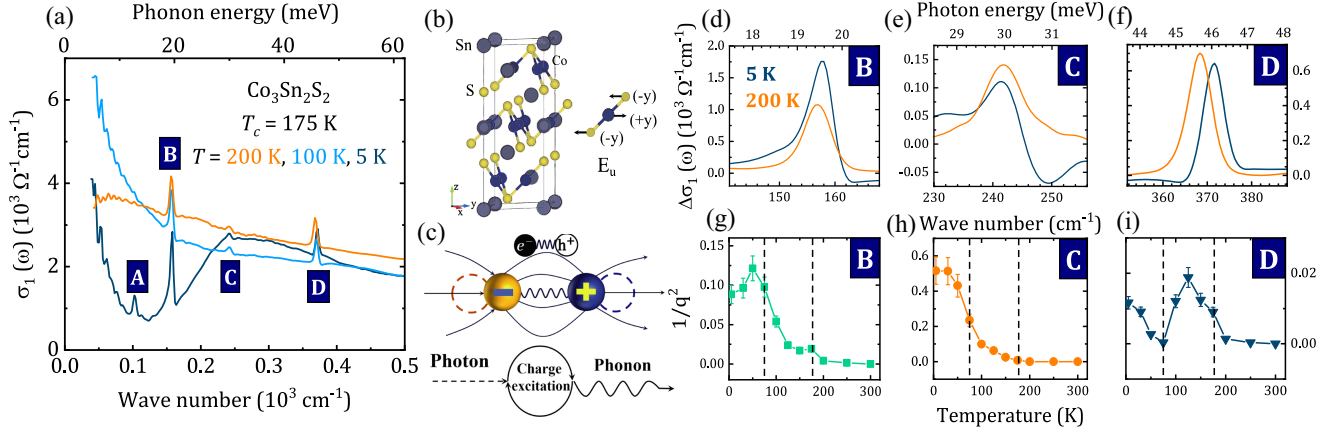


FIG. 1. Electron-phonon coupling below the Curie temperature. (a) Optical conductivity $\sigma_1(\omega)$ of $\text{Co}_3\text{Sn}_2\text{S}_2$ below 62 meV. The infrared-active phonons are marked by capital letters. (b) Lattice structure (left) and the vibration of E_u mode (right). (c) The cartoon (upper) and the diagram (lower) describing the EPC. (d)–(f) The phonon line shapes below and above the Curie temperature (175 K). The electronic background is subtracted by polynomial fitting in narrow energy range. (g)–(i) Fano factors ($1/q^2$) acquired by fit the phonon line shapes with Fano mode.

between phonon and electronic systems can result in asymmetric Fano line shape of the phonon absorption peaks in the optical conductivity, $\sigma_1(\omega)$ [40–42]. On the other side, since circularly polarized phonons absorb right and left circularly polarized light differently [19], when the T of phonon field is broken, discernible CD can be reflected by finite polar Kerr rotation angle and ellipticity [33,43,44].

In this work, with the reflectivity and polar Kerr rotation spectroscopy [$\theta_k(\omega)$] measurements, we studied the dynamic behaviors of the infrared-active phonons in magnetic Weyl semimetal $\text{Co}_3\text{Sn}_2\text{S}_2$, which has a centrosymmetric lattice structure [Fig. 1(b)]. Below the T_c , $\text{Co}_3\text{Sn}_2\text{S}_2$ has six symmetry-related Weyl nodal rings without considering the spin-orbit coupling (SOC) and becomes a Weyl semimetal after considering the SOC. In the far-infrared energy range, we observed asymmetric absorption peaks of the E_u mode phonons, which has been regarded as the signature of the EPC. The asymmetry of the phonon line shapes become more prominent upon cooling, which suggests an increase in the coupling strength between phonons and charge excitations on the tilted Weyl nodal rings. Meanwhile, the polar Kerr rotation and ellipticity spectroscopy detected significant CD of these phonons. By analyzing the asymmetry of the phonon line shapes in circularly polarized optical conductivities [$\sigma_1^\pm(\omega)$, $+/ -$ represent right- and left-hand channels], we find that the phonons couple differently with the electronic topology for two circular polarizations. Such unequal EPC in two polarizations breaks the T of phonon field, giving rise to the splitting of E_u mode. Moreover, since the magnetism gradually modulates the topological states, the EPC as well as the CD of the phonons also exhibits a noticeable temperature (T) dependence.

Electron-phonon coupling below the T_c —Figure 1(a) displays $\sigma_1(\omega)$ above and below the T_c (see Sec. I of

Supplemental Material (SM) [45] and Ref. [46] for details of the measurements). The optical conductivity is dominated by the smooth and continuous responses from the absorptions for charge excitations [46,56], while several sharp peaks (marked by A, B, C, and D) stem from the absorptions of infrared-active phonons. Based on the first-principles calculations, these sharp peaks can be attributed to the E_u mode vibrations (Fig. S5 of SM [45]). In this Letter, we will focus on the behavior of these phonons. Upon cooling, along with the dramatic change of charge excitations, the phonon peaks exhibit a remarkable T dependence. In addition to the usual sharpening, their line shapes become asymmetric as T is reduced. In Figs. 1(d)–1(f), after subtracting the electronic background using quadratic fit in short energy ranges, it becomes evident that the phonon line shapes exhibit a pronounced asymmetry at low T . Such an asymmetric line shape has been observed in graphene [47], superconductors [40,57], and Weyl semimetals [41], and attributed to the EPC, during which the discrete phonon excitations have comparable energies with the continuous charge excitations and interfere with each other [42]. The stronger EPC results in more asymmetric line shape. Since phonon A is screened by itinerant carriers above 75 K and exhibits a symmetric line shape [Fig. 1(a)], we will concentrate on the properties of phonons B, C, and D (the reasons are discussed in Sec. II of SM [45]).

The asymmetry of the phonon peaks can be quantitatively described by the Fano model,

$$\sigma_1(\omega) = \frac{2\pi\Omega^2 q^2 + \frac{4q(\omega-\omega_0)}{\gamma} - 1}{Z_0 \gamma q^2 \left(1 + \frac{4(\omega-\omega_0)^2}{\gamma^2}\right)}, \quad (1)$$

in which Z_0 is the vacuum impedance; ω_0 , γ , and Ω correspond to the resonance frequency, linewidth, and

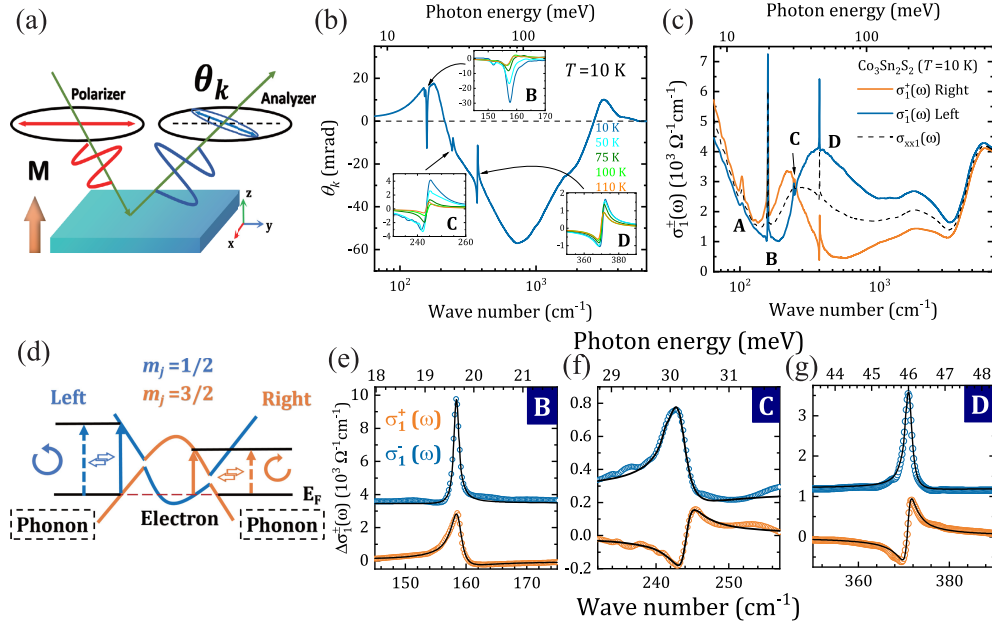


FIG. 2. Circular dichroism of the phonons in $\text{Co}_3\text{Sn}_2\text{S}_2$. (a) Sketch of the polar Kerr rotation measurement, during which the light is parallel to the magnetic moment but perpendicular to the sample surface. (b) Kerr rotation angle spectroscopy below 0.8 eV measured at $T = 10$ K. The insets depict the Kerr rotation spectroscopy of phonons B, C, and D measured at various T , in which the electronic background is subtracted. (c) Circularly polarized optical conductivity $\sigma_1^\pm(\omega)$ of $\text{Co}_3\text{Sn}_2\text{S}_2$ at 10 K. The four infrared-active phonons are marked by the capital letters. The dashed line represents the unpolarized optical conductivity $\sigma_{xx}(\omega) = 1/2[\sigma_1^+(\omega) + \sigma_1^-(\omega)]$. (d) Sketch of EPC in each circular polarization, in which the bands constituting the Weyl nodal rings harbor different angular momenta of $+3/2$ (orange) and $+1/2$ (blue), respectively. The solid and dashed lines represent the direct interband transitions for phonons and electrons. (e)–(g) Line shapes of phonons B, C, and D, in each circular polarization at 10 K. The black solid lines are fits by Fano model.

strength of the phonon, respectively; the Fano factor $1/q^2$ or $1/q$ quantifies the asymmetry, with a higher value indicating a more pronounced asymmetric line shape (see Sec. II of SM [45] for details). The T dependence of $1/q^2$ displayed in Figs. 1(g)–1(i) reveals that the asymmetry of the phonons B, C, and D emerges right below $T_C \approx 175$ K and becomes more apparent at lower T , suggesting a crucial role played by magnetism in driving the EPC in $\text{Co}_3\text{Sn}_2\text{S}_2$. While the asymmetry of the phonons B and C increases continuously upon cooling, the phonon D asymmetry shows a nonmonotonic T dependence and reaches its minimum at 75 K, indicating two distinct EPC processes, as discussed later [58].

At low T , the electronic background in $\sigma_1(\omega)$ below 0.1 eV was attributed to the charge excitations on tilted Weyl nodal rings [43,46]. Thus, for comparable energies, phonons B, C, and D primarily interfere with the charge excitations within the topological states [59]. SOC leads to distinct angular momenta of the bands constituting Weyl nodal rings [as shown in Fig. 2(d) and Fig. S7]. For the conservation of angular momentum, charge excitations on different sides of the Weyl nodal rings should absorb photons with different chiralities [43]. However, the Weyl nodal rings in $\text{Co}_3\text{Sn}_2\text{S}_2$ are tilted, for the Pauli blockade effect [60], charge excitations on the Weyl nodal rings absorb the photons with different energies in each

circular polarization [sketched in Fig. 2(d)]. Because phonons couple to charge excitations, it is intriguing whether the electronic system can transmit the CD to phonons through EPC in centrosymmetric $\text{Co}_3\text{Sn}_2\text{S}_2$.

Circular dichroism of the phonons in $\text{Co}_3\text{Sn}_2\text{S}_2$ —With the optical configuration in Fig. 2(a), we measured the polar Kerr rotation and ellipticity of $\text{Co}_3\text{Sn}_2\text{S}_2$ (see Sec. I of SM [45] and Ref. [44] for the details of the measurements). The polar Kerr effect arises from the difference in absorption of light with opposite circular polarizations [43,44]. From the results shown in Fig. 2(b), one can see considerable Kerr rotation angle (θ_k) in broad energy ranges at 10 K [the ellipticity (η_k) is shown in Fig. S1 [45]]. At 90 meV, θ_k is as large as 60 mrad, such strong Kerr rotation was also observed by Y. Okamura *et al.* and attributed to the presence of tilted Weyl nodal rings near the Fermi level [43]. Below 0.1 eV, both θ_k and η_k change signs, indicating that the charge excitations of different energies on the tilted Weyl nodal rings have distinct preferences for one of the polarizations [61] (the discussion about the electronic background is given in Sec. I of SM [45]). In addition to the electronic responses, we also observed several sharp peaks in $\theta_k(\omega)$ and $\eta_k(\omega)$ (Fig. S1 [45]); these sharp peaks occur at the same energies as the phonons B, C, and D, hinting at the CD of phonon absorptions [33,37]. (The reason for the absent signal of phonon A is discussed

in Sec. II of SM [45]). With increasing T , the signals of these phonons gradually diminish [insets of Fig. 2(b)].

In order to investigate the origin of CD in $\text{Co}_3\text{Sn}_2\text{S}_2$, we derived $\sigma_1^\pm(\omega)$ based on the reflectivity and the Kerr rotation spectra [Fig. 2(c), refer to Sec. I of SM [45] for details]. As expected, at 10 K $\sigma_1^\pm(\omega)$ exhibit a significant disparity in the two circular polarizations. The absorption peak at 30 meV in unpolarized optical conductivity [represented by the dashed curve in Fig. 2(c)] was attributed to charge excitations on tilted Weyl nodal rings [46]. As mentioned above, the charge excitations on different sides of the tilted Weyl nodal rings absorb photons differently for the two circular polarizations [Fig. 2(d)]; thus, the 30 meV absorption peak splits with different energies for $\sigma_1^+(\omega)$ and $\sigma_1^-(\omega)$ [depicted by the blue and orange curves in Fig. 2(c) [62]; the corresponding $\sigma_{xy1}(\omega)$ is calculated in Fig. S1(d) and discussed in Sec. I of SM [45]].

In $\sigma_1^\pm(\omega)$, after subtracting the electronic background, the phonon absorptions still show asymmetric line shapes. However, the line shapes for opposite polarizations are rather distinct for phonons B, C, and D, indicating different EPC [Figs. 2(e)–2(g)]. The phonon peaks are fitted with the Fano model [black lines in Figs. 2(e)–2(g) and S4]. Figures 3(a), 3(c), and 3(e) clearly demonstrate that $1/q^2$ in the right-hand channel is significantly larger for phonons B, C, and D. The distinct EPC also results in splitting of the phonon positions in two polarizations. For phonons B and C, the center frequencies are higher in the right-hand channel, and the splitting between two circular polarizations increases as T is lowered [Figs. 3(b) and 3(d)], while the splitting of phonon peak D exhibits a nonmonotonic evolution that even reverses across 75 K [Fig. 3(f)]. Since $\sigma_1(\omega)$ is the average of the $\sigma_1^\pm(\omega)$, the reversal of phonon D splitting leads to the nonmonotonic evolution of its asymmetry in Fig. 1(i). The broken degeneracy of phonons imparts them with finite angular momentum [8].

Discussion—So far, optical spectroscopy has revealed significant T -dependent EPC in magnetic Weyl semimetal $\text{Co}_3\text{Sn}_2\text{S}_2$ below T_C . Since the energies of the infrared-active phonons overlap with those of charge excitations on the tilted Weyl nodal rings, the discrete phonons mainly interfere with the continuum of the topological electronic state. Because of the SOC and Pauli blockade effect, charge excitations on the tilted Weyl nodal rings exhibit remarkable CD. A close inspection of $\sigma_1^\pm(\omega)$ reveals that the phonons B, C, and D couple differently with the charge excitations in two circular polarizations [Figs. 2(e)–2(g)].

The asymmetric phonon line shape observed by optical spectroscopy is known as the charged-phonon effect: the phonon modes borrow dipole intensities from the electronic background [Fig. 1(c)] [42]. In $\text{Co}_3\text{Sn}_2\text{S}_2$, the infrared-active phonons stem from linear dipole vibrations (E_u mode), which can be regarded as the degeneracy of left- and right-handed circularly polarized modes. As phonons

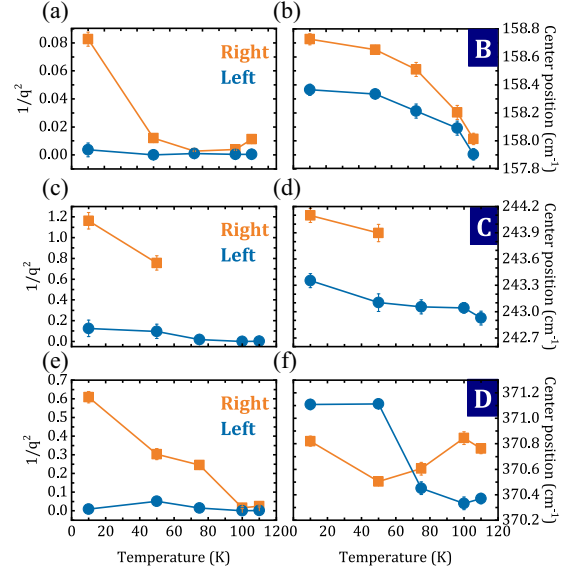


FIG. 3. Temperature dependence of the parameters used to fit the phonon modes B, C, and D. (a),(c),(e) Temperature dependence of the Fano factor $1/q^2$ in each circular polarization. (b),(d), (f) Temperature dependence of peak positions in each circular polarization.

exchange energies with the charge excitations distinctly in two circular polarizations, the degeneracy of phonons will be lifted. Consequently, the linear vibrations transform into elliptical ones.

Moreover, the phonon-peak splitting in the two circular polarizations exhibit an interesting T dependence. For phonons B and C, the center frequency is always higher in the right-hand channel, with separation expanding upon cooling [Figs. 3(b) and 3(d)]. The splitting of phonon D, however is rather different: when passing 75 K, it reverses the direction [Fig. 3(f)]. From theoretical considerations, it was pointed out recently that in a T -breaking system, the molecular Berry curvature (MBC) induced by the coupling between phonons and electronic topology can break the degeneracy of optical phonons at the Γ point, resulting in phonon splitting into right- and left-hand polarized branches. The direction and magnitude of the splitting are determined by the real part of MBC at the Γ point [7,8,39]. In Ref. [8], EPC only happens in one circular polarization since the valleys are polarized, leading to a single splitting direction at the Γ point. However, the tilted Weyl nodal rings in $\text{Co}_3\text{Sn}_2\text{S}_2$ are formed by band crossing and therefore EPC occurs in both circular polarizations [Fig. 2(d)]; the resulting phonon splittings are opposite and compete with each other [7]. Thus, the MBC should be considered in two circular polarizations. According to its definition [Eq. (17) of Ref. [8]], MBC at the Γ point corresponds to the phonon-induced direct interband transitions [8], which can be evaluated by the strength of EPC and the charge excitation probability. The EPC can be quantized by the Fano factor $1/q$, while the probability of

charge excitation is proportional to the electronic background in the optical conductivity $\sigma_1^{e\pm}(\omega_0)$.

In Fig. S8 of SM [45], we plot the values of $|1/q \times \sigma_1^{e\pm}(\omega_0)|$, which corresponds to the MBC in opposite circular polarizations. For phonons B and C, both the EPC and the charge excitations at the phonon frequencies ω_0 are always stronger in right-hand polarization, resulting in larger MBC and higher phonon frequencies [Figs. 3(b) and 3(d)]. For phonon D, at $T < 75$ K, even though the EPC is more intense in the right-hand polarization, the charge excitations are much stronger in the left-hand polarization [Fig. 2(c)], giving rise to larger MBC and higher phonon frequencies for that polarization [Fig. 3(f)]. Since $\text{Co}_3\text{Sn}_2\text{S}_2$ is an itinerant ferromagnet, with increasing T , the magnetic exchange gradually shifts the Weyl nodal rings away from the Fermi level [46,63]. With the suppressed charge excitations on the Weyl nodal rings and EPC in both circular polarizations, the splitting of the B and C phonons diminishes. However, at the frequency of phonon D, the gap in charge excitations between two circular polarizations narrows at high T (Fig. S9), then the larger EPC gives rise to larger MBC in right-hand polarization and reverses the splitting across 75 K. The correspondence between $|1/q \times \sigma_1^{e\pm}(\omega_0)|$ and phonon splitting further reflects that the circular phonons stem from coupling of phonons to the topological electronic states.

The reversal of phonon D splitting is reminiscent of the reversed phonon magnetic momentum or g factor predicted [7] and observed in $\text{Pb}_{1-x}\text{Sn}_x\text{Te}$ [33]; this phenomenon was attributed to the band inversion during the topological phase transitions. Nevertheless, in these situations, EPC only occurs in one circular polarization and an external magnetic field is required to break the $\mathcal{T}$ of phonon field. In $\text{Co}_3\text{Sn}_2\text{S}_2$, the EPC exists in both polarizations. Because of the competition of two splitting effects, the Pauli blockade effect, and magnetism modification to the topological states, the properties of circular polarized phonons are especially rich here, even in the absence of an external field. In particular, the T -dependent phonon splitting in opposite circular polarizations provides an efficient way to manipulate the phonon polarization. Since the magnetic exchange effect shifts the topological state [46,63] and the spin flip reverses the Kerr rotations of both charge and phonons (Fig. S10 of SM [45]), manipulating the phonon polarization through doping [64], gate voltage [65], and an external magnetic field—besides T —holds promise for future studies.

Summary—Using reflectivity and polar Kerr rotation spectroscopy measurements, we observed evidence of circularly polarized phonons in centrosymmetric magnetic Weyl semimetal $\text{Co}_3\text{Sn}_2\text{S}_2$ without an external magnetic field. By analyzing the asymmetric phonon line shapes in the optical conductivity, we attribute the CD of phonons to the unequal couplings between phonons and electronic topology in opposite circular polarizations. As magnetism

modifies the topological bands continuously, the CD of the phonons exhibits a pronounced T dependence. Thus, this proof-of-principle study establishes a new avenue for exploring circularly polarized phonons, and provides a more flexible approach to manipulate the circular polarization and even the chirality of phonons.

Note added—We noticed that recently the circular dichroism of phonons in $\text{Co}_3\text{Sn}_2\text{S}_2$ has also been confirmed by Raman scattering measurement [66].

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